

Manipulations de fractions

Fiche n°4.1

TEMPS ESTIME : 15 MIN

Mettre les expressions sous la forme : $\frac{S}{E} = \frac{a_0 + a_1x + a_2x^2 + \dots + a_nx^m}{x^\alpha(1 + b_1x + b_2x^2 + \dots + b_nx^n)}$

1	$2Sx + 4ax(E + bSx^2) + Scx = 0$
2	$S = bx(E(3x + 2) + 4(S - E)) + Scx^2$
3	$S = \frac{K}{3x} \left[\frac{1}{-x^2 - x + 2} (S - 2.E) - \frac{1}{x^2 + x - 2} E \right]$
4	$\frac{S}{E} = \frac{\frac{ax + 4}{3x + c} \left[\frac{x^2 + b}{x} + \frac{3x + 4}{2} \right]}{\frac{dx + 1}{3x + c}}$

Manipulations de fractions

Fiche n°4.2

TEMPS ESTIME : 15 MIN

Mettre les expressions sous la forme : $\frac{S}{E} = \frac{a_0 + a_1x + a_2x^2 + \dots + a_nx^m}{x^\alpha(1 + b_1x + b_2x^2 + \dots + b_nx^n)}$

1	$cS(x + 2) + \frac{cS(x + 2)^2}{-3x} - \frac{E(6x)}{(4x - 2)} = 0$
2	$\frac{S}{E} = \frac{\frac{ax^2 + 3x + 4x}{2x} \left[\frac{x^2 + \frac{b}{4x}}{x} - c \frac{x + 4}{2} \right]}{\frac{d}{2x}}$
3	$S = 2Ex + 4E(2 - 3ax) + \frac{1}{1 + bx + cx^2} (S - 4E)$
4	$S = \frac{1}{(1 + x)} \frac{1}{(1 + 2x)^2} [S - (1 + x)E]$

①

$$2Sx + 4ax(E + bSx^2) + Scx = 0$$

 $\Leftrightarrow$

$$2Sx + 4axE + 4abxSx^2 + Scx = 0$$

 $\Leftrightarrow$

$$S(2x + 4abx^2 + c) = -4axE$$

 $\Leftrightarrow$

$$\frac{S}{E} = \frac{-4ax}{2x + 4abx^2 + c}$$

 $\Leftrightarrow$

$$\frac{S}{E} = \frac{-4ax}{(2+c)x + 4abx^2}$$

$$\frac{S}{E} = \frac{-4a}{(2+c) + 4abx}$$

$$\frac{S}{E} = \frac{-4a}{2+c} \cdot \frac{1}{1 + \frac{4abx}{2+c}}$$

②

$$S = bx(E(3x+2) + 4(S-E)) + Scx^2$$

$$\Leftrightarrow S = bx(3Ex + 2E + 4S - 4E) + Scx^2$$

$$\Leftrightarrow S = bx(3Ex - 2E + 4S) + Scx^2$$

$$\Leftrightarrow S = 3Ebx^2 - 2Ebx + 4Sbx + Scx^2$$

$$\Leftrightarrow 0 = E(3bx^2 - 2bx) + S(4bx + cx^2 - 1)$$

$$\Leftrightarrow E(3bx^2 - 2bx) = -S(4bx + cx^2 - 1)$$

$$\Leftrightarrow \frac{S}{E} = -\frac{3bx^2 - 2bx}{4bx + cx^2 - 1}$$

$$\Leftrightarrow \frac{S}{E} = \frac{3bx^2 - 2bx}{1 - 4bx - cx^2}$$

③

$$S = \frac{K}{3x} \left[\frac{1}{-x^2 - x + 2} (S - 2E) - \frac{1}{x^2 + x - 2} E \right]$$

$$S = \frac{K}{3x} \left[\frac{1}{-x^2 - x + 2} S - \frac{2}{-x^2 - x + 2} E - \frac{1}{x^2 + x - 2} E \right]$$

$$S = \frac{K}{3(-x^3 - x^2 + 2x)} S - \frac{2K}{3(-x^3 - x^2 + 2x)} E - \frac{K}{3(x^3 + x^2 - 2x)} E$$

$$S \left(1 - \frac{K}{3(-x^3 - x^2 + 2x)} \right) = E \left(-\frac{2K}{3(-x^3 - x^2 + 2x)} - \frac{K}{3(x^3 + x^2 - 2x)} \right)$$

$$\frac{S}{E} = \frac{-\frac{2K}{3(-x^3 - x^2 + 2x)} + \frac{K}{3(x^3 + x^2 - 2x)}}{1 - \frac{K}{3(-x^3 - x^2 + 2x)}}$$

$$\frac{S}{E} = \frac{-K}{3(-x^3 - x^2 + 2x) - K}$$

$$\frac{S}{E} = \frac{-K}{3(-x^3 - x^2 + 2x) - K} \times \frac{3(-x^3 - x^2 + 2x)}{3(-x^3 - x^2 + 2x) - K}$$

$$\frac{S}{E} = \frac{-K}{3x(-x^2 - x + 2) - K} = -\frac{K}{-3x^3 - 3x^2 + 6x - K}$$

$$= \frac{1}{1 - \frac{6}{K}x + \frac{3}{K}x^2 + \frac{3}{K}x^3}$$

$$\textcircled{4} \quad \frac{S}{E} = \frac{\frac{ax+4}{3x+c} \left[\frac{x^2+b}{x} + \frac{3x+4}{2} \right]}{\frac{dx+1}{3x+c}}$$

$$\frac{S}{E} = \frac{\frac{ax+4}{3x+c} \left[\frac{2(x^2+b) + x(3x+4)}{2x} \right]}{\frac{dx+1}{3x+c}}$$

$$\frac{S}{E} = \frac{(ax+4)(2(x^2+b) + x(3x+4))}{2x(dx+1)}$$

$\times (3x+c)2x$ En haut et en bas

$$\frac{S}{E} = \frac{(ax+4)(2x^2+2b+3x^2+4x)}{2dx^2+2x}$$

$$\frac{S}{E} = \frac{(ax+4)(5x^2+4x+2b)}{2dx^2+2x}$$

$$\frac{S}{E} = \frac{5x^3a + 4x^2a + 2xb + 10x^2 + 16x + 8b}{2dx^2+2x}$$

$$\frac{S}{E} = \frac{8b + (2ba+16)x + (4a+10)x^2 + 5ax^3}{x(2dx+2)}$$

$$\frac{S}{E} = \frac{4b + (ba+8)x + (2a+10)x^2 + \frac{5}{2}ax^3}{x(dx+1)}$$

①